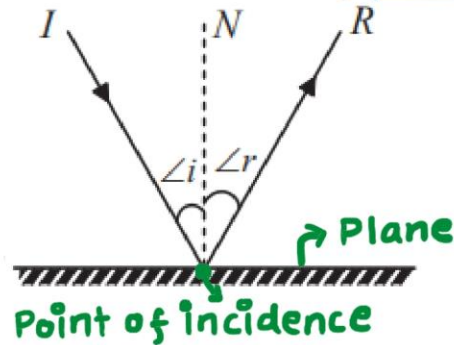


LIGHT: REFLECTION AND REFRACTION

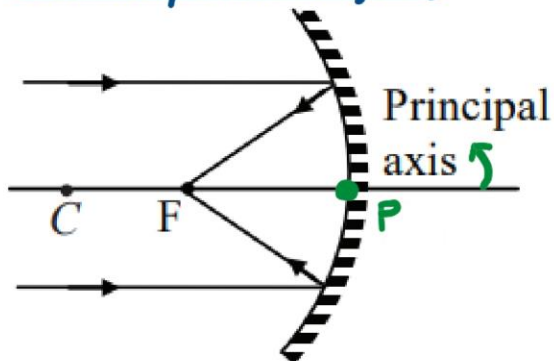
REFLECTION OF LIGHT AND ITS LAWS

- **Reflection**: bounce back of light when it hits any **polished surface** like mirror
- **Laws**:
 - 1 $\angle i = \angle r$
 - 2 Incident ray, reflected ray & normal lies in same plane



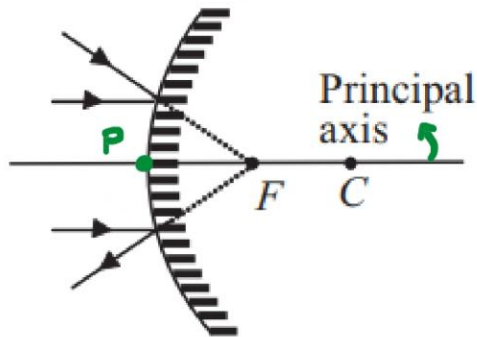
SPHERICAL MIRRORS

concave/ converging



- **Pole (P)**: midpoint on mirror surface
- **Centre of curvature (C)**: centre of sphere from which mirror is cut

convex/ diverging

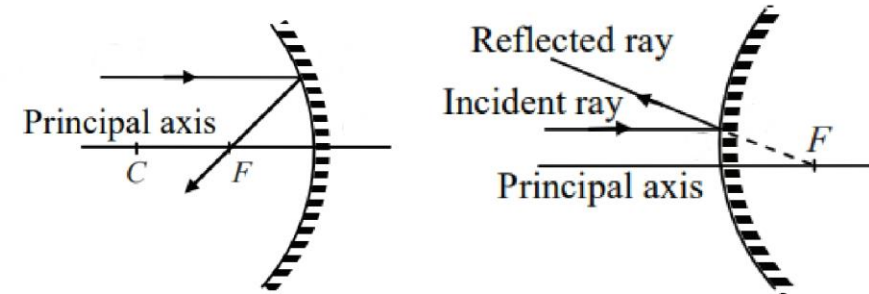


- **Principal axis**: joins 'P' and 'C'
- **Radius of curvature**: distance b/w 'P' and 'C'
- **Focal length (f)**: distance b/w 'P' and 'F' (Focus)
NOTE: 'F' is centre of 'P' and 'C' $\therefore R = 2f$

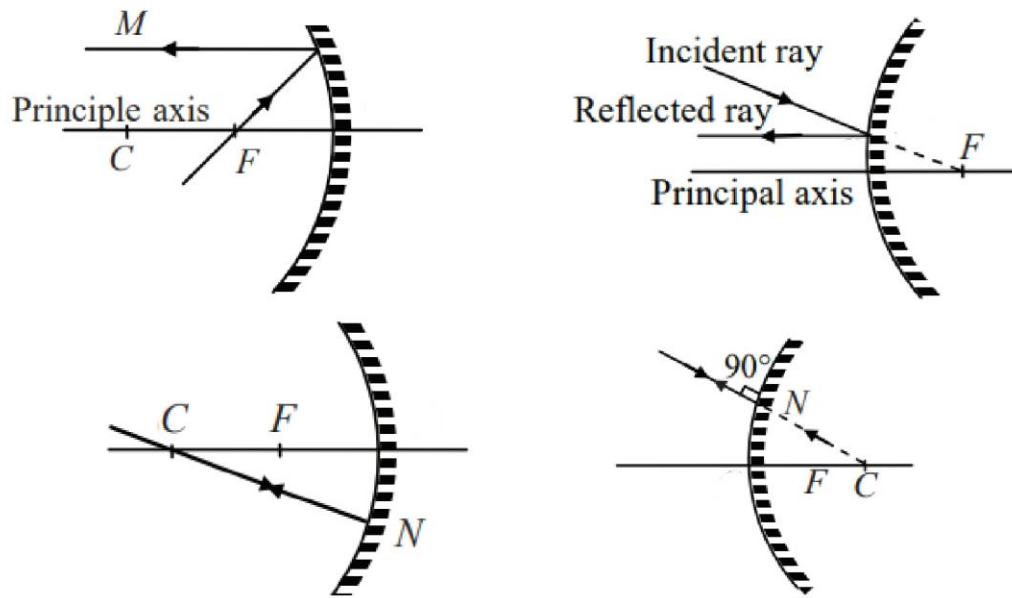
IMAGE FORMATION: RULES

- At least **2 rays** have to meet to form an image
- If rays actually meet: **Real image** is formed
appear to meet: **Virtual image** is formed

- 1 Rays parallel to principal axis, pass through 'F'



- 2 Rays passing through 'F' will be parallel to principal axis
- 3 Rays passing through 'C' will reflect back along the same path



4 When a ray incident at 'P' : $\angle i = \angle r$; here, principal axis behaves as normal (N)

IMAGE FORMATION BY CONCAVE MIRROR

u (position of object)	v (position of image)	Diagram	Nature of image
At infinity	At 'F'		Real; inverted; point sized (highly diminished)

• NOTE: Real images are inverted ; virtual ones are erect

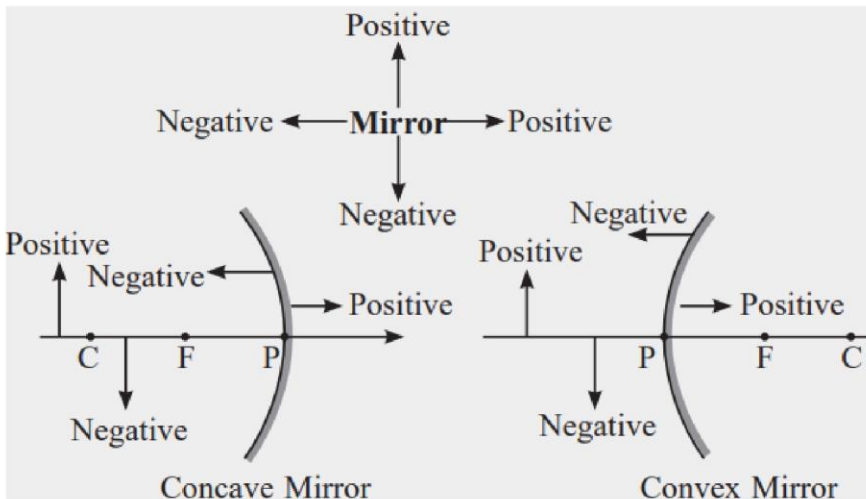
Beyond 'C'	Between 'F' and 'C'		Real; inverted and diminished
At 'C'	At 'C'		Real; inverted; same sized
Between 'C' and 'F'	Beyond 'C'		Real; inverted; bigger than the object
At 'F'	At infinity		Real; inverted; highly enlarged
Between 'P' and 'F'	Behind the mirror		Virtual; erect; enlarged; behind the mirror

IMAGE FORMATION BY CONVEX MIRROR

u (position of object)	v (position of image)	Diagram	Nature of image
At infinity	At 'F'		Virtual; erect; point sized; behind the mirror
Between infinity and 'P'	Between 'P' and 'F'		Virtual; erect; diminished; behind the mirror

SIGN CONVENTION

Note: All distances measured from 'P'



- u = always $\ominus$ ve
- convex = f = $\oplus$ ve
- concave = f = $\ominus$ ve
- Mirror formula:

$$\frac{1}{f} = \frac{1}{u} + \frac{1}{v}$$

MAGNIFICATION

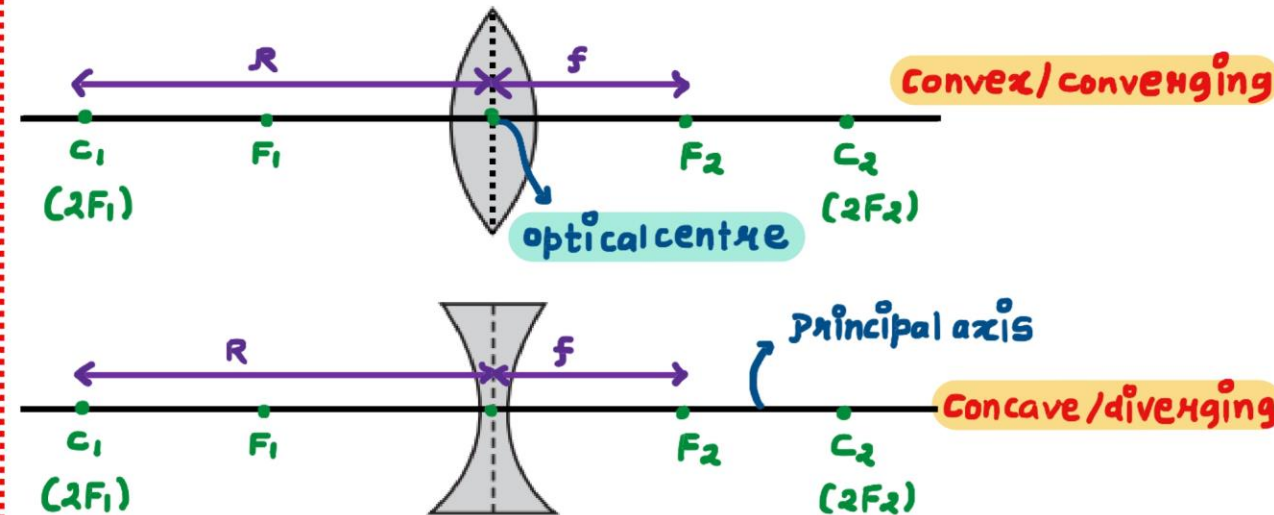
- Tells how large or small an image is w.r.t object
- Magnification = $\frac{\text{height of image}}{\text{height of object}} = \frac{h_i}{h_o} = -\frac{v}{u}$

Note: m = $\ominus$ ve : real image ; m = $\oplus$ ve : virtual image

USES OF MIRRORS

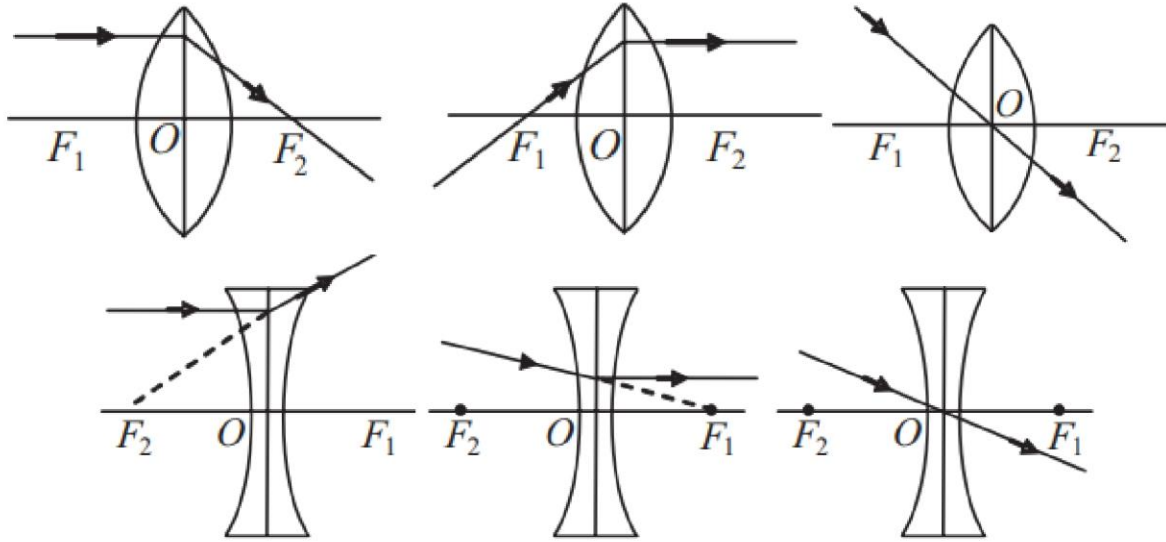
CONCAVE MIRROR	CONVEX MIRROR
Torch, headlights, shaving mirror, dentist mirror (enlarged image)	Real view mirrors (erect and small image and hence wide view)

SPHERICAL LENSES



• C: centre of curvature ; F: principal focus ; f: focal length

IMAGE FORMATION: RULES



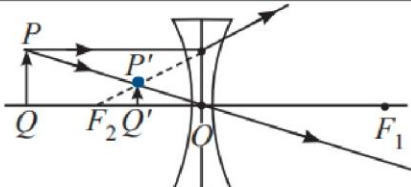
At ' $2F_1$ '	At ' $2F_2$ '		Real; inverted; same sized
Between ' F_1 ' and ' $2F_1$ '	Beyond ' $2F_2$ '		Real; inverted; enlarged
At ' F_1 '	At infinity		Real; inverted; highly enlarged
Between ' F_1 ' and '0'	Same side of the lens as object		Virtual, erect and enlarged

IMAGE FORMATION BY CONVEX LENS

u (position of object)	v (position of image)	Diagram	Nature of image
At infinity	At ' F_2 '		Real; inverted; highly diminished (point sized)
Beyond ' $2F_1$ '	Between ' F_2 ' and ' $2F_2$ '		Real; inverted and diminished

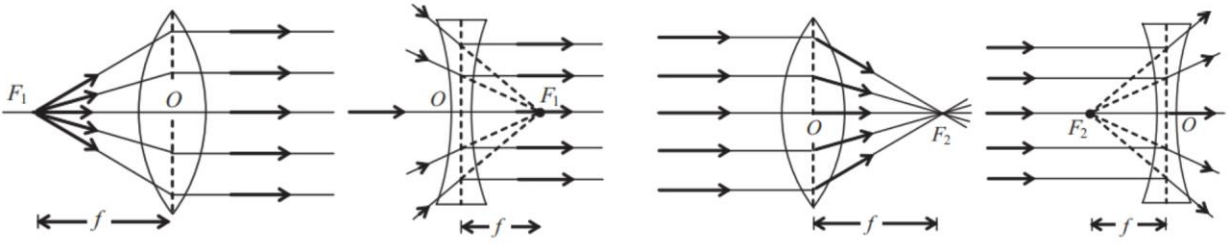
IMAGE FORMATION BY CONCAVE LENS

u (object)	v (image)	Diagram	Nature of image
At infinity	At ' F_1 '		Virtual, erect and highly diminished (point sized)

Between 'infinity' and '0'	Between 'F1' and '0'		Virtual, erect and Diminished
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IMPORTANT NOTES

• Spherical lens have 2 focus



• Sign convential : all distances measured from 'O'

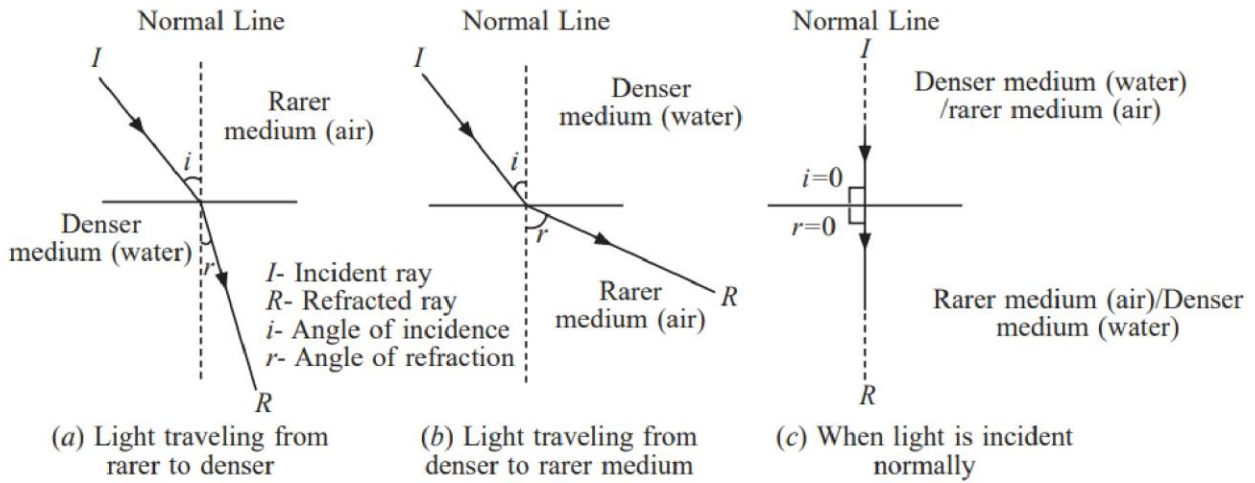
• $\frac{1}{f} = \frac{1}{v} - \frac{1}{u}$; $m = \frac{h_i}{h_o} = \frac{v}{u}$; signs same as mirror

• Power of lens: Degree of convergence & divergence of a lens ; SI unit : Dioptre (D) = m^{-1} $\therefore P = 1/f$

• Power: concave : -ve ; convex : +ve

REFRACTION: BENDING OF LIGHT

• It occurs due to change in medium which leads to change in speed of light



• Refractive index (RI): tells about density of a medium
RI of 2nd medium w.r.t 1

$n_{12} = \frac{n_2}{n_1} = \frac{v_1}{v_2}$; RI $\uparrow$; density $\uparrow$; speed of light

• Absolute refractive index: one medium is air ; $n=1$
 $v = c = 3 \times 10^8 m/s$

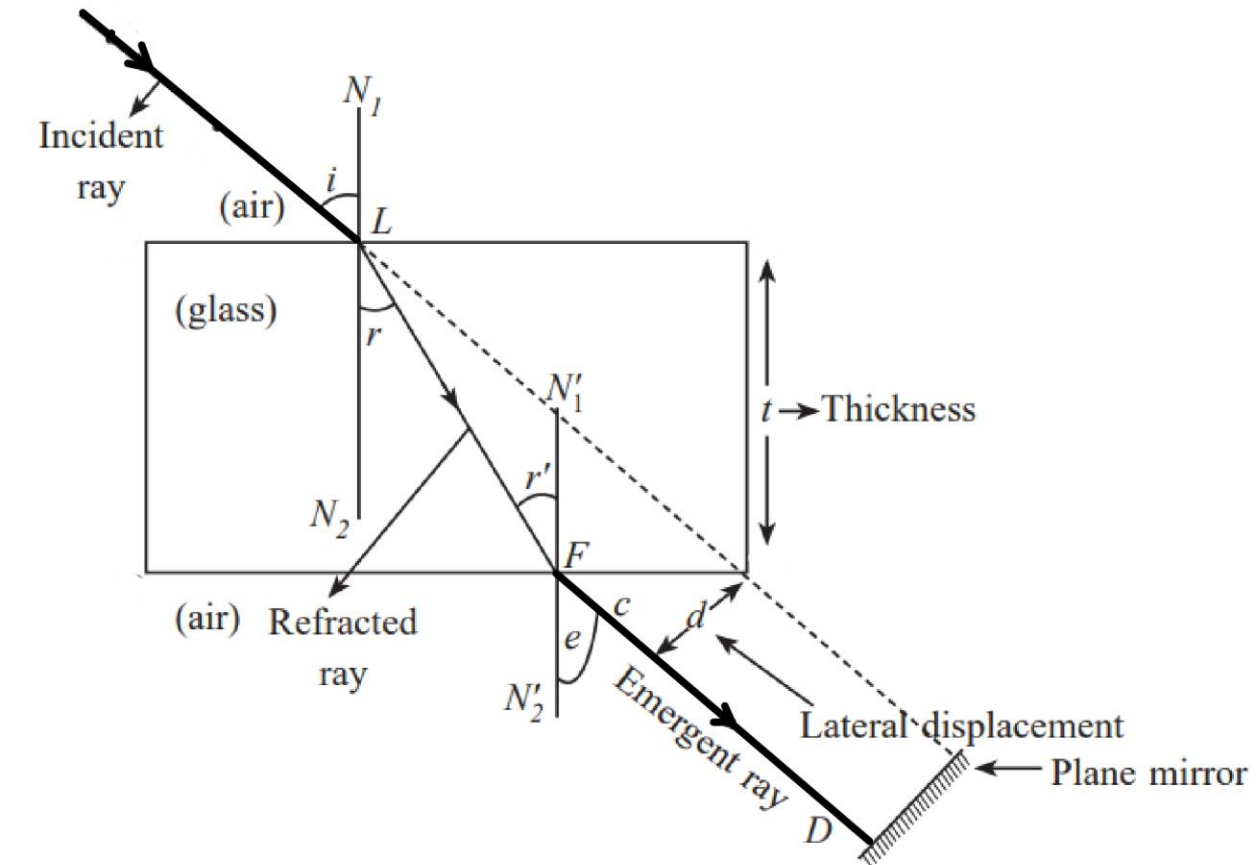
• 'n' of glass = 1.5 ; 'n' of water = 1.33

• Laws of refraction:

- 1 Incident ray, refracted ray & normal lie in same plane
- 2 Ratio of sin of incidence ray with normal & sin of refracted ray with normal remains constant for a given pair of media

$\therefore$ if we change $\angle i$; $\angle r$ also changes but ratio remained same

REFRACTION THROUGH A GLASS SLAB



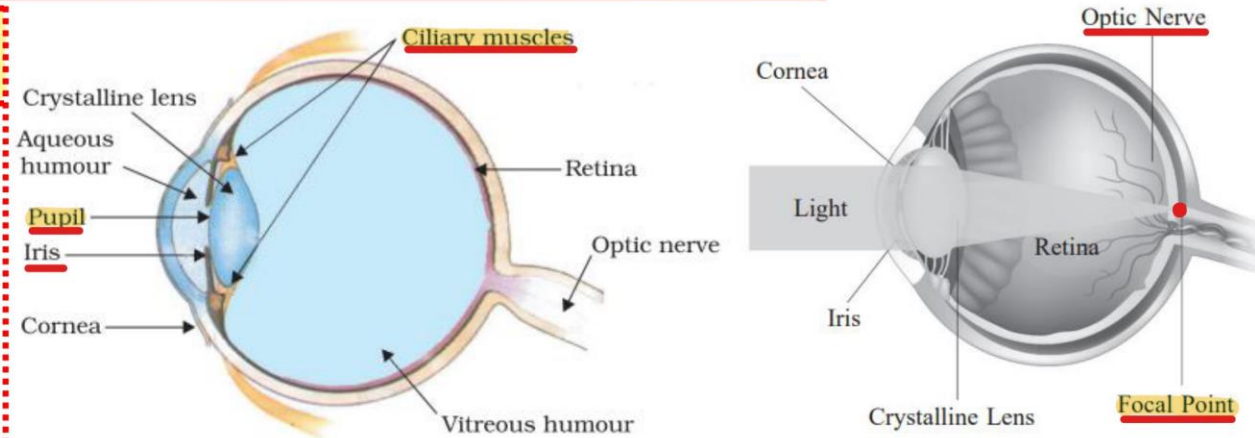
• Incident ray is parallel to emergent ray

• $\angle i = \angle e$

HUMAN EYE AND THE COLOURFUL WORLD

HUMAN EYE

Part of Eye	Features
Cornea	Thin, transparent, outer membrane which refracts the light entering the eye
Iris	Dark bigger circle seen in eye that controls the size of pupil
Pupil	Small circle seen at centre of iris which controls the amount entering the light
Crystalline lens	Convex; adjusts focal length to see different objects; make real and inverted image
Ciliary muscles	Changes curvature of eye lens and thus controls focal length
Retina	Screen where image is formed; it contains sensitive cells which generates electrical signals when light falls on them Rods (low light vision); Cones (Coloured, high light vis.)
Optic nerve	Takes signal to brain in form of electric signal
Aqueous humour (water like)	Between cornea and lens; nutritive and supportive
Vitreous humour (gel like)	Between lens and retina; supportive



POWER OF ACCOMODATION

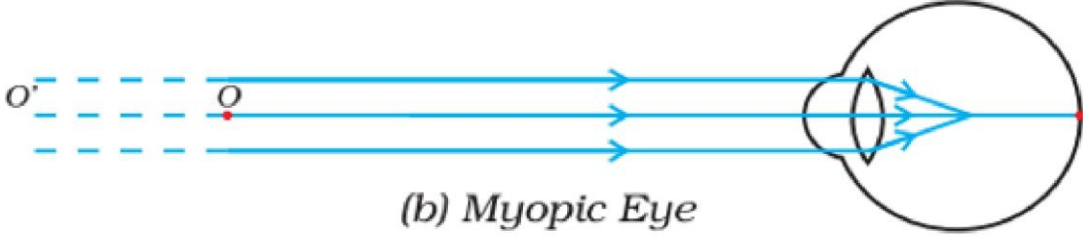
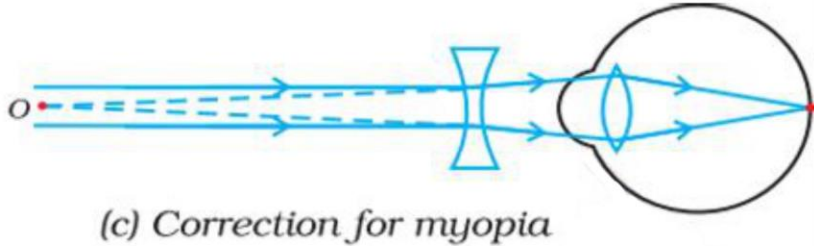
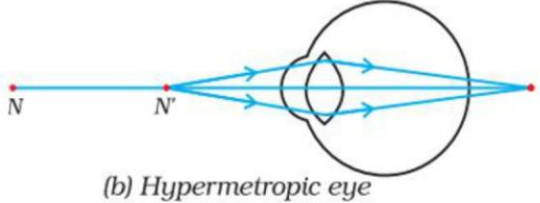
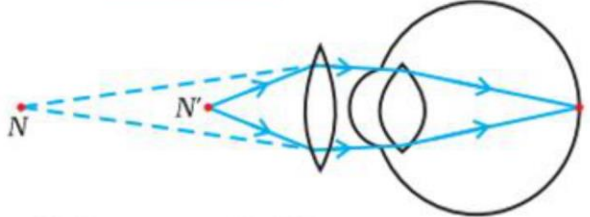
- Ability of eye to change its focal length
- To see closer objects: ciliary muscles contract
 - ∴ lens gets thick ∴ curvature ↑
 - ∴ focal length ↓ ∴ Power ↑ ∴ $P = 1/f$
- To see farther objects: ciliary muscles relax;
 - ∴ lens gets thin ∴ curvature ↓
 - ∴ focal length ↑ ∴ power ↓

NEAR AND FAR POINT

- Farthest point upto which eye can see (∞)

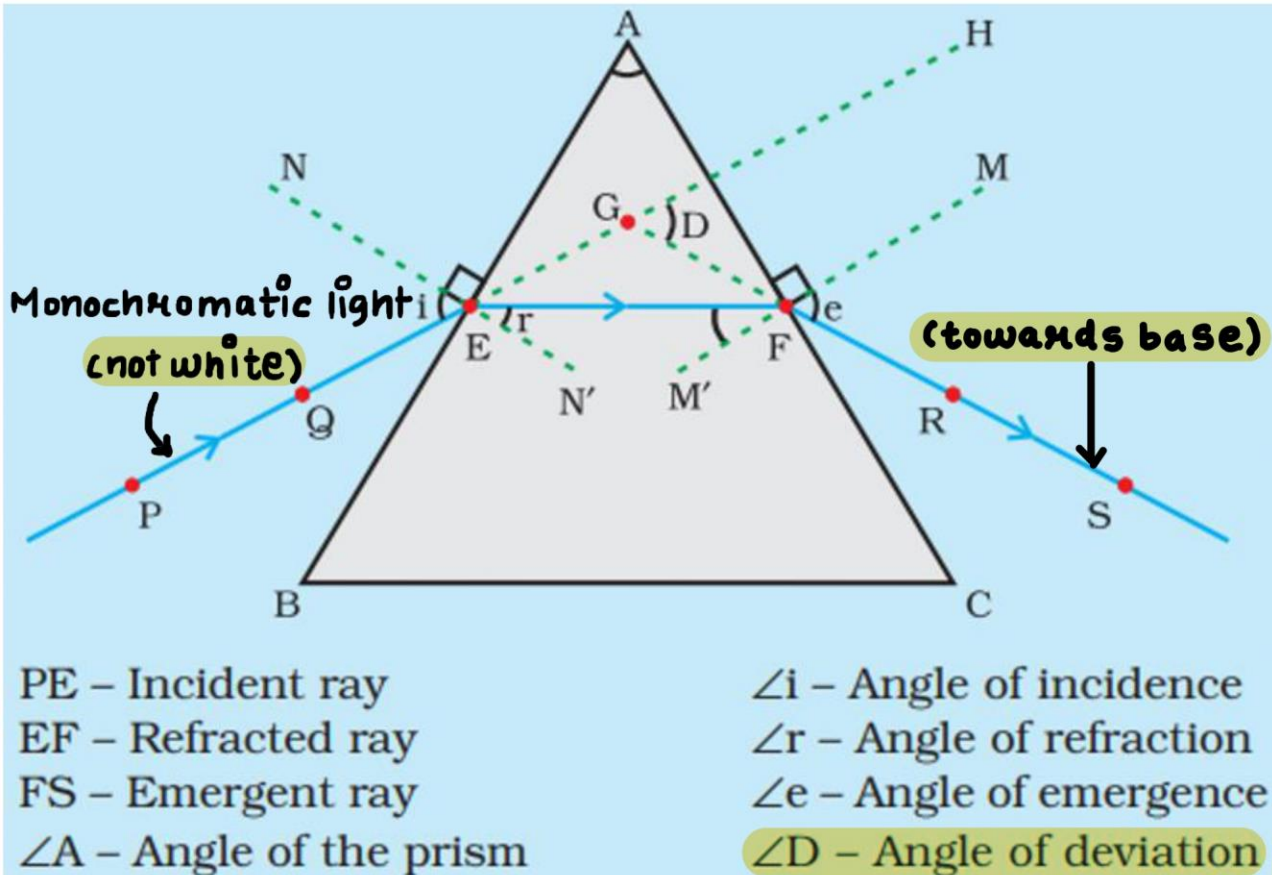
- Near point or Least Distance of Distinct Vision (LDDV): Minimum distance at which objects can be seen without strain (25cm)

DEFECTS OF VISION AND THEIR CORRECTION

Defects	Reasons	Corrections
Myopia (Near sightedness) • Their far point is less than infinity	• Excessive curvature of eye lens (thick) and thus focal length decrease • image formed before retina since eyeball enlarges  (b) Myopic Eye	• Use of concave lens of suitable power  (c) Correction for myopia
Hypermetropia (Far sightedness) • Their near point is more than 25cm	• Lens become thin and thus focal length increases; • Eyeball becomes small and thus image is made behind retina  (b) Hypermetropic eye	• Use of convex lens of suitable power  (c) Correction for Hypermetropic eye
Presbyopia • Unable to see both near and far objects • Age related	• Weakening of ciliary muscles and decreased flexibility of lens	• Use of bifocal lenses 
Cataract	Lens become opaque/ milky which leads to partial or complete loss of vision	• Surgery

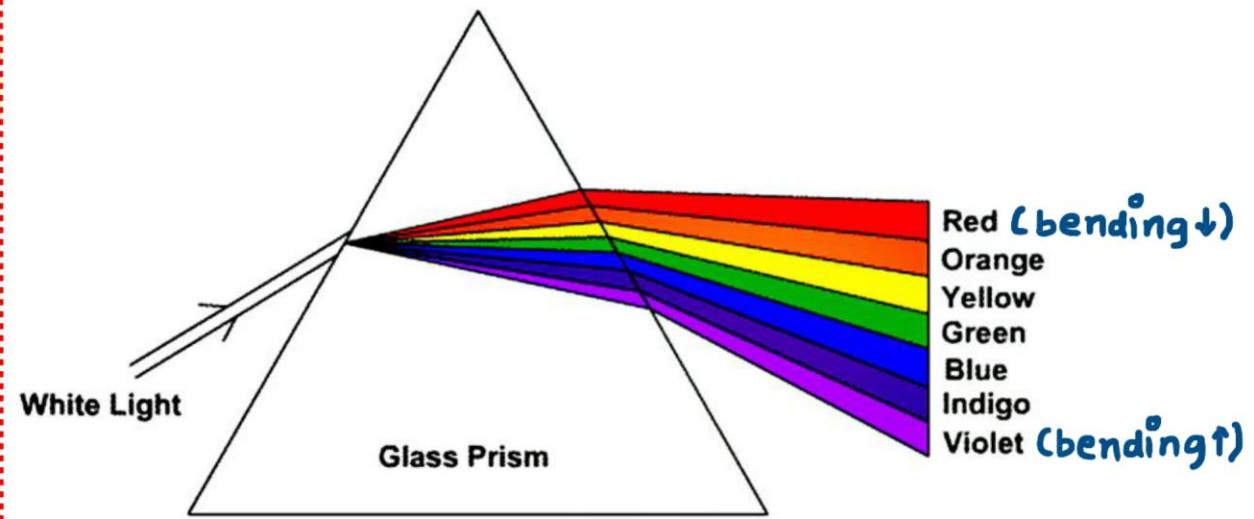
REFRACTION THROUGH A PRISM

- Prism is a glass which has 2 triangular & 3 rectangular surfaces $\therefore$ it is a transparent refracting medium

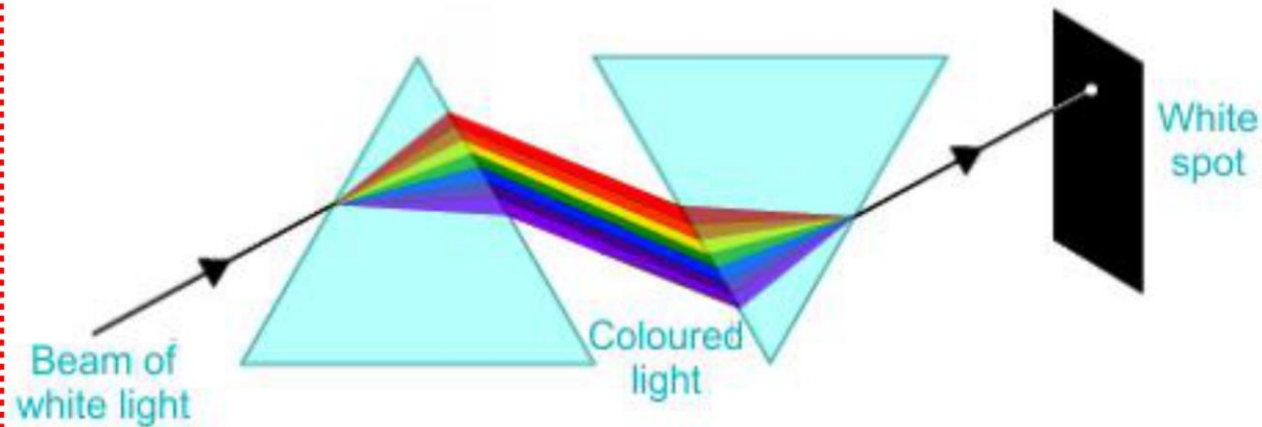


- When white light/sunlight strike on prism, it splits in 7 colours (violet bends the most: $\angle D \uparrow$); red bends least

SPLITTING/ DISPERSION

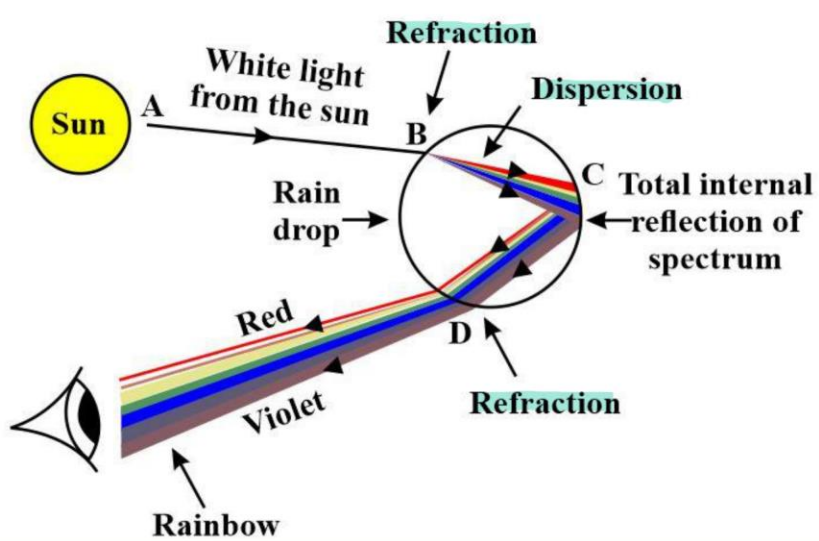


NEWTON'S INVERTED PRISM EXPERIMENT



RAINBOW FORMATION

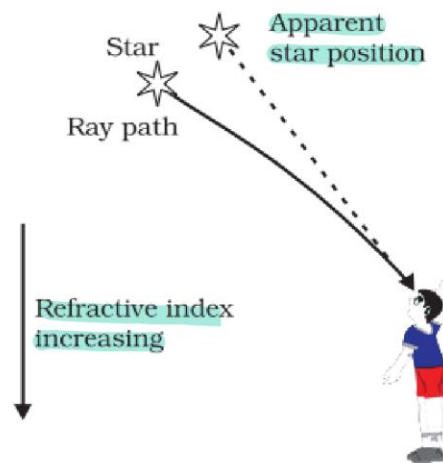
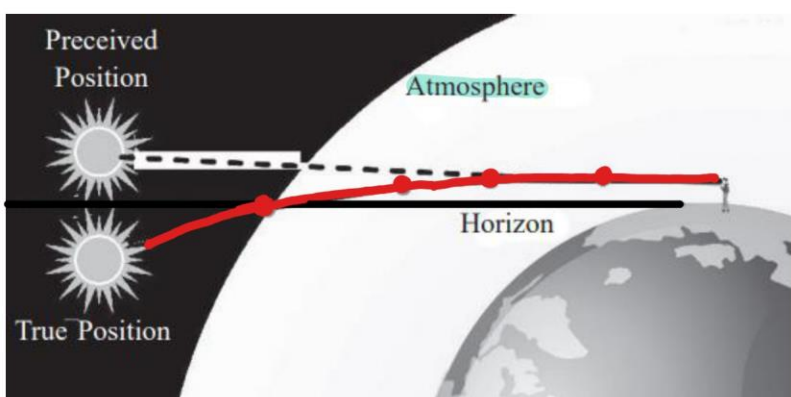
- Due to refraction, dispersion & total internal reflection
- Sun should be behind us & rainbow is formed in front



ATMOSPHERIC REFRACTION

- Different layers of atmosphere has different densities
- Layers close to earth have higher densities

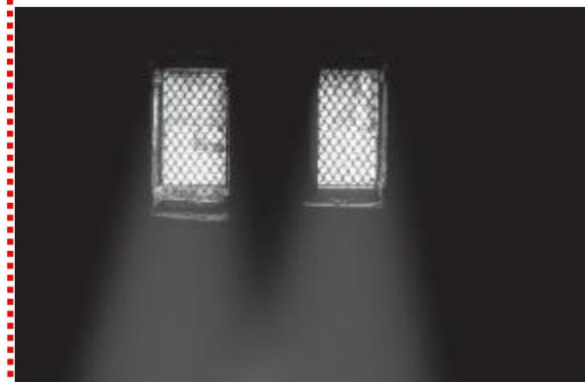
- 1 Apparent position of stars
- 2 Advance sunrise & delayed sun-set



TWINKLING OF STARS

- Stars are point source of light; Physical condition of diff. atm. layers keeps on changing due to change in temp., pressure & densities $\therefore$ The path of light coming from stars changes $\therefore$ no. of rays/amount of light entering our eye flickers/varies $\therefore$ Star appears twinkling
- Planets do not twinkle: they are near to earth $\therefore$ behaves as extended source of light (collection of million of point sized sources) $\therefore$ nullify twinkling

TYNDALL EFFECT/ SCATTERING OF LIGHT



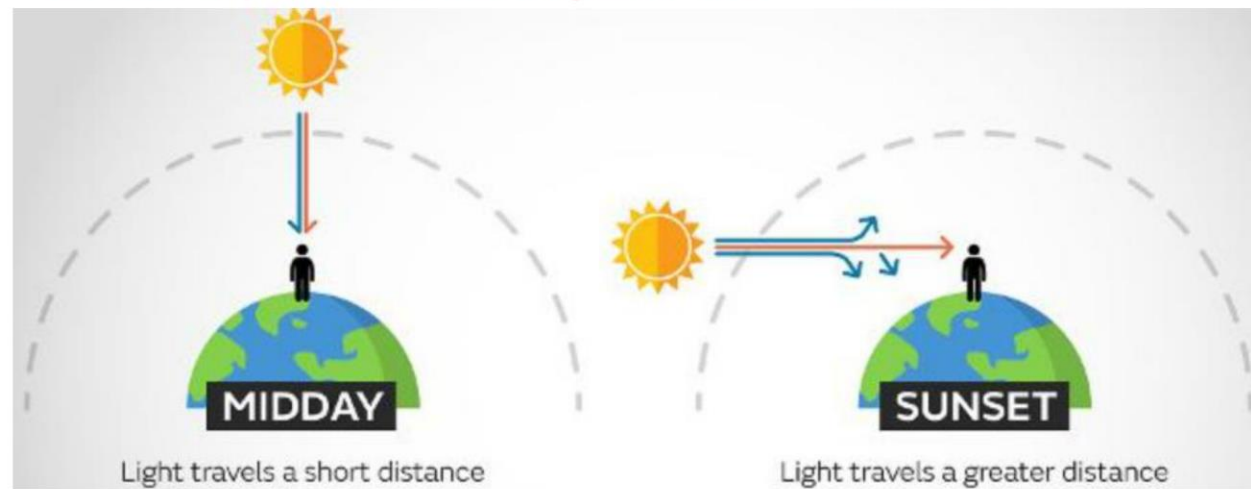
- Scattering of light by heterogeneous colloidal particles
- e.g., our environment has smoke, H_2O , dust, air particles
- colour of scattered light depends on size of particles
- e.g., very tiny particle: Blue ($\lambda \downarrow$); Large particles: Red; very large particles (white colour)

- 1 Blue colour of sky: molecules of air & other fine particles scatter blue light strongly than red



Note: If no atmosphere, then no scattering $\therefore$ sky dark

- 2 sun appear red during sunrise & sunset



- 3 Danger & Stop signs are usually red

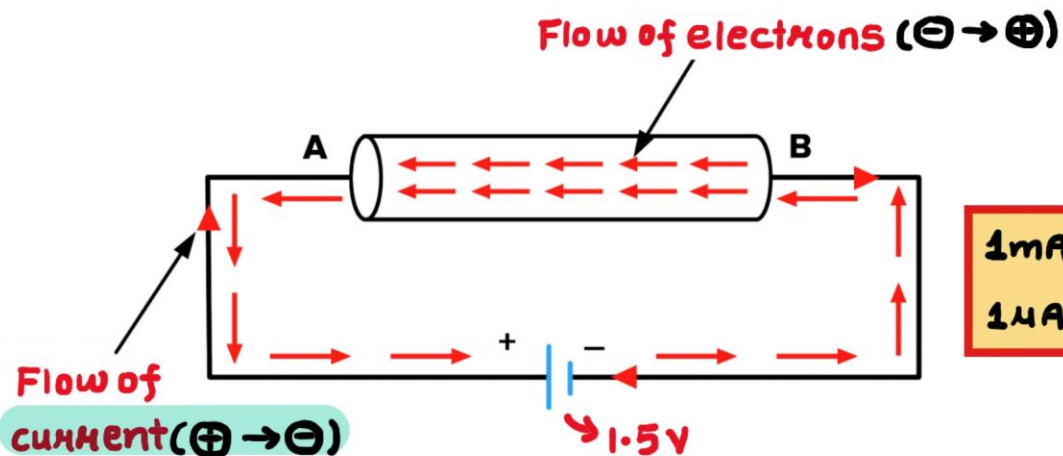


Particles of fog & smoke scatter red colour least ($\lambda \uparrow$) $\therefore$ visible from large distance

ELECTRICITY

CHARGE (Q) AND CURRENT (I)

- **Charge**: opposite attract; like charges repel; unit-coulomb (C); $1C = 6 \times 10^{18}$ electrons $\therefore$ charge on an electron is $1.6 \times 10^{-19}C$ ($\therefore e^-$ is smallest independent charge)
- **Electric current (I)**: Amount of charge (Q) flowing through any cross section of conductor in time (t)
 $\therefore$ It is rate of flow of charges $\therefore I = Q/t$ or $Q = It$
Units: I = Ampere; Q = coulomb; t = second
- **1 Ampere** = flow of '1C' charge in '1sec' = C/s or Cs^{-1}
- **Ammeter**: Instrument used to measure electric current



$$1mA = 10^{-3}A$$
$$1\mu A = 10^{-6}A$$

POTENTIAL DIFFERENCE (V)

- **Potential**: work done to move a charge from one place to another
 $\therefore V = \frac{W}{Q}$; V in volt, W in joule, Q in coulomb
- **1 volt**: 1J work done to move 1C charge from one place to another $\therefore 1V = 1J C^{-1}$
- **Potential difference**: Diff. of potential b/w 2 points in an electric circuit which is produced by chemical reaction across terminal of cell; measured by volt meter
- Pot. diff. is necessary for the current flow

RESISTANCE (R)

- Property of conductor to resist flow of charges through it i.e., **obstruction**; Unit = ohm (Ω)
- **1 'ohm'**: Flow of 1A current through conductor b/w 2 points of 1V potential difference $\therefore 1\Omega = 1V/1A$
- **NOTE**: when e^- moves, they are obstructed

$V = IR \therefore I \propto \frac{1}{R}$ $\therefore$ If R doubled, then I halves

• Factors affecting resistance:

① Nature of material: good or bad conductors

② Length of conductor: $R \propto l$

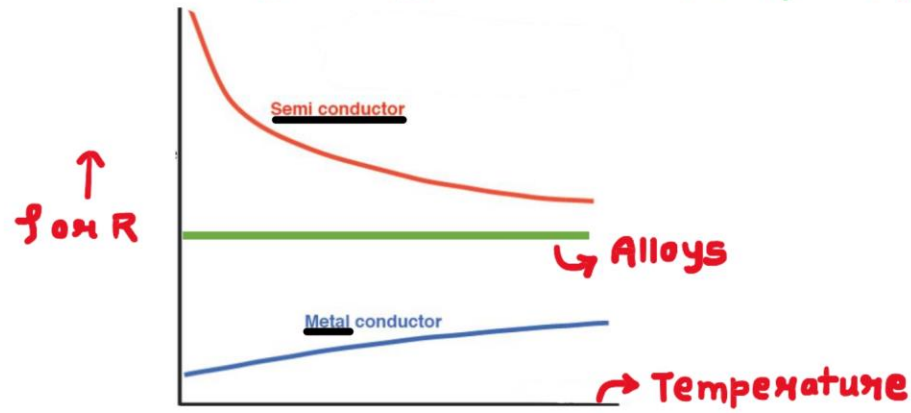
③ Area of cross-section: $R \propto \frac{1}{A}$

$\therefore R \propto \frac{l}{A} \therefore R = \frac{\rho l}{A}$; unit of ' ρ ' = Ωm

• Resistivity (ρ): characteristic property of a material
 $\hookrightarrow$ resistance offered by a substance of unit length
 & unit area of cross-section

• ρ and $R \downarrow$ ($\rho = 10^{-8}$ to $10^{-6} \Omega m$): conductor: metal, alloys

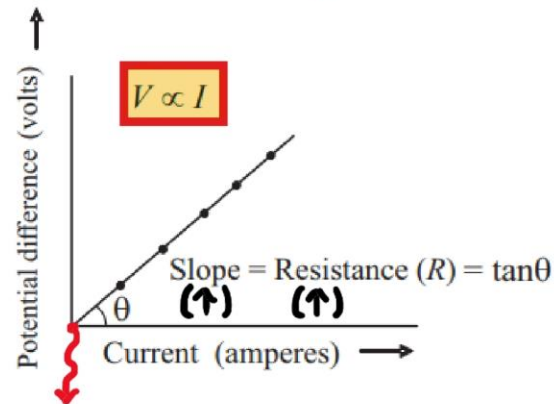
• ρ and $R \uparrow$ ($\rho = 10^{12}$ to $10^{17} \Omega m$): Insulator: Rubber, glass, plastic



- $R = \frac{\rho l}{A}$; if radius ' r ' is given, use $R = \frac{\rho l}{\pi r^2}$
- If diameter is given, calculate ' r ' first (in metres)

OHM'S LAW

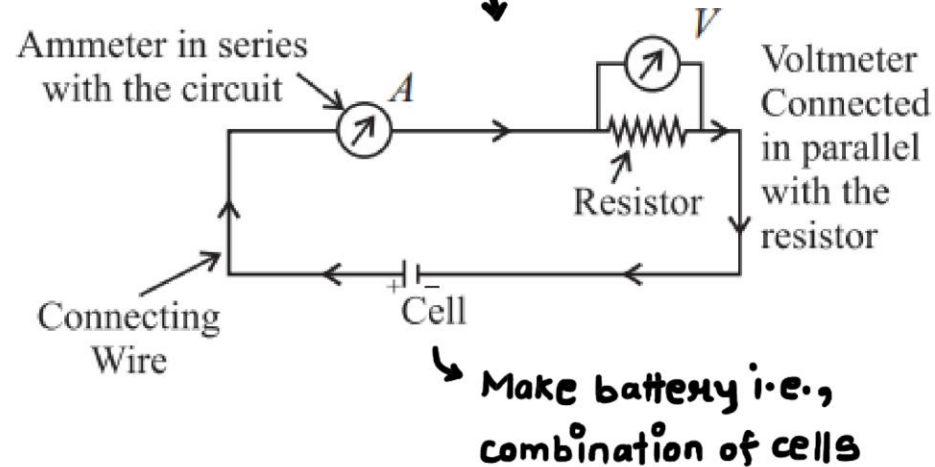
- Pot. diff. (V) across ends of a metallic conductor is directly proportional to current (I) flowing through it (provided Temp. remains same)



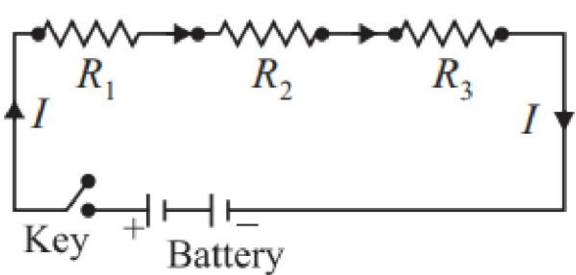
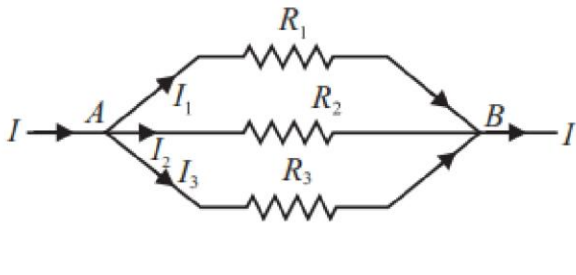
If $V=0, I=0$

Experimental Setup

Increase battery voltage step by step & take ammeter reading at each step

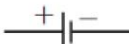
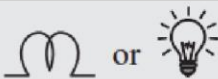
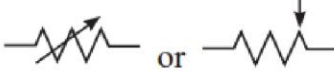
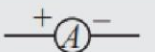
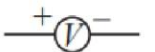


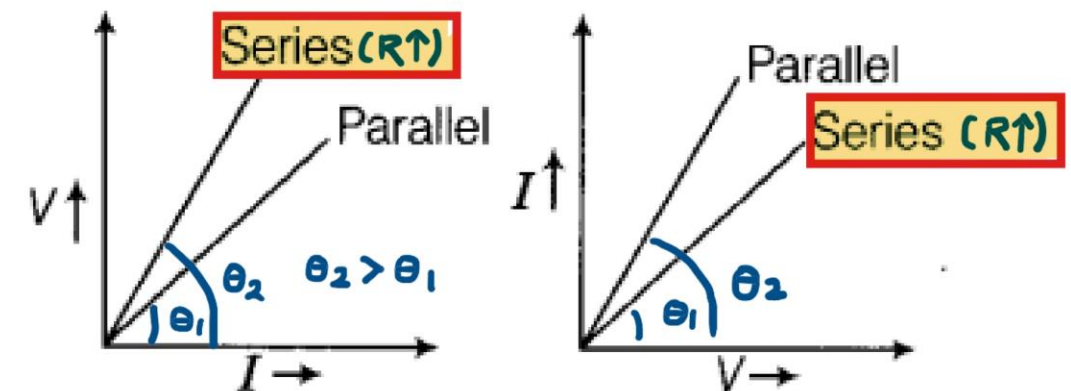
COMBINATION OF RESISTORS

Series Combination	Parallel Combination
	
Current is same in all resistors 'V' is different i.e., IR_1 , IR_2 and IR_3	Current is different but 'V' is same between 2 points
$R_e = R_1 + R_2 + R_3$	$\frac{1}{R_e} = \frac{1}{R_1} + \frac{1}{R_2} + \frac{1}{R_3}$
Max resistance i.e., $R_e > R_1, R_2, R_3$	Minimum resistance: $R_e < R_1, R_2, R_3$
Disadvantages: - If one component fails then current don't pass furthers (circuit break) - Electric bulb and toaster cannot be connected in series as they need different amount of current - Overall resistance is high	Advantages: - Every component gets equal voltage - Current is divided to different components - Easy to connect and disconnect new element without breaking the circuit - Overall resistance is less

CIRCUIT DIAGRAM

- Circuit is continuous closed path of electric current

Components	Symbols
Electric cell	
Battery	
A resistor of resistance R	
Electric bulb	
Variable resistance or rheostat	
Plug key (switch open) - OFF	
Plug key (switch closed) - ON	
Ammeter	
Voltmeter	
A wire joint	
Wires crossing without joining	
Fuse	



ELECTRIC POWER (P)

- Rate at which electrical energy is consumed

$$P = VI \rightarrow \text{Ampere}$$

↙ volt
↘ Watt

$$\therefore V = IR$$

$$P = I^2 R \quad \text{or} \quad P = \frac{V^2}{R}$$

ELECTRICAL ENERGY (E)

- Supplied by the cell; SI unit = Joule

$$E = Pt \rightarrow \text{Hour}$$

↙ Kilowatt
↘ kWh

$$1 \text{ kWh} = 3.6 \times 10^6 \text{ J}$$

↳ 1 commercial unit of Bijli Bill

HEATING EFFECT OF ELECTRIC CURRENT

- If an electric circuit is purely resistive, the source energy gets dissipated entirely in form of heat
- Used in Iron, oven, kettle, heater etc.
- Joules law of heating: $H \propto I^2$

$$H \propto R$$

$$H \propto t$$

(time for which current flows) ↘

$$\therefore H = I^2 R t$$

- H in Joule; 'I' in ampere; R in ohm & 't' in seconds

Alloys are used here $\therefore \uparrow \text{resistivity} \therefore H = I^2 R t$

$\therefore \uparrow \text{m.pt}$

$\therefore \text{donot oxidise}$

ELECTRIC BULB

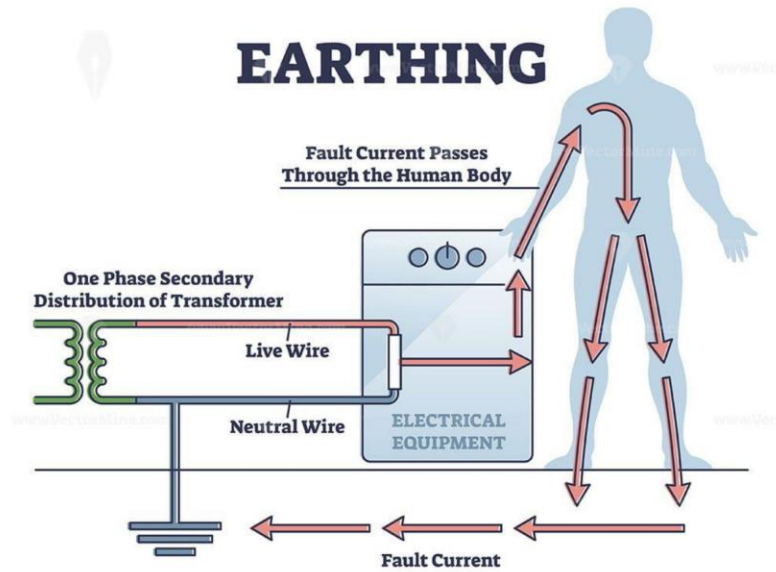
- Filament made up of tungsten (m.pt. = 3380°C); filled with N_2 & argon to increase life of filament

ELECTRIC FUSE

- Its a safety device; prevents short circuit & overloading
- When high current pass through it, it melts & stops the flow of current
- It is made of: Al, Cu, Pb, Fe (low m.pt): usually Sn coated Cu is used (Tin coated copper)
- It should have $\uparrow \therefore R \uparrow \therefore \text{Heat} \uparrow \therefore \text{melt}$
- If 4.5 A current flows through an appliance the 5A fuse must be used
- 15 ampere fuse in iron, geyser & 5A in bulb, fan

EARTHING OF ELECTRICAL APPLIANCE

- **Safety measure**
- Discharges powerful surges to soil safely (avoid accidents)
- Earth wire connected to metallic part of appliances so that if current leaks, it can go down to earth



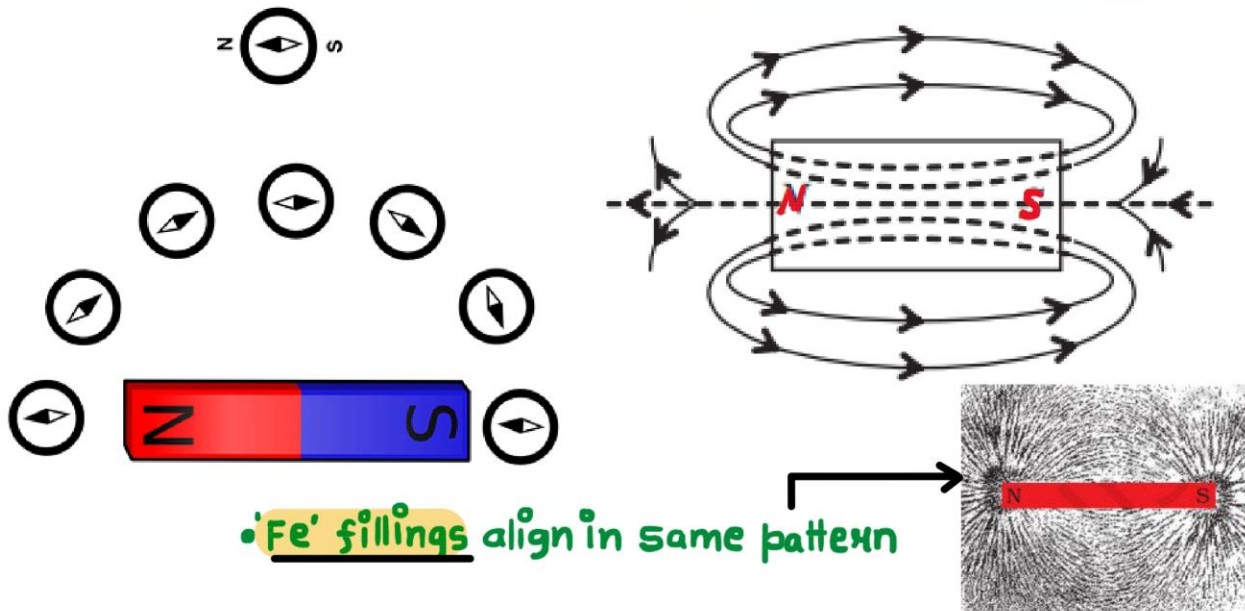
MAGNETIC EFFECTS OF CURRENT

MAGNET AND COMPASS

- Attract opp. pole of other magnet or metals like Fe, Cobalt, Nickel
- A small needle shaped magnet that deflects due to effect of magnet
Mag-field $\uparrow$, then deflection $\uparrow$

MAGNETIC FIELD (B)

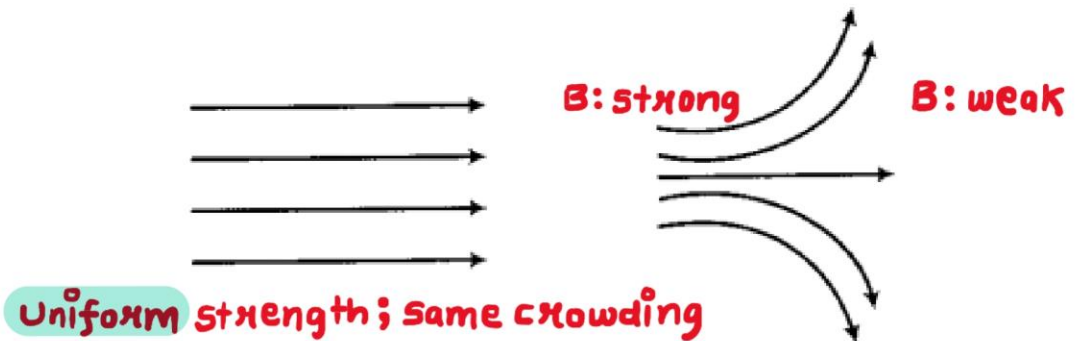
- Area around a magnet where magnetic force is experienced; represented by magnetic field lines



Properties of Magnetic field:

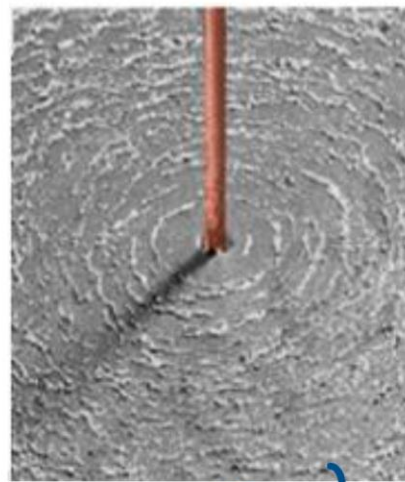
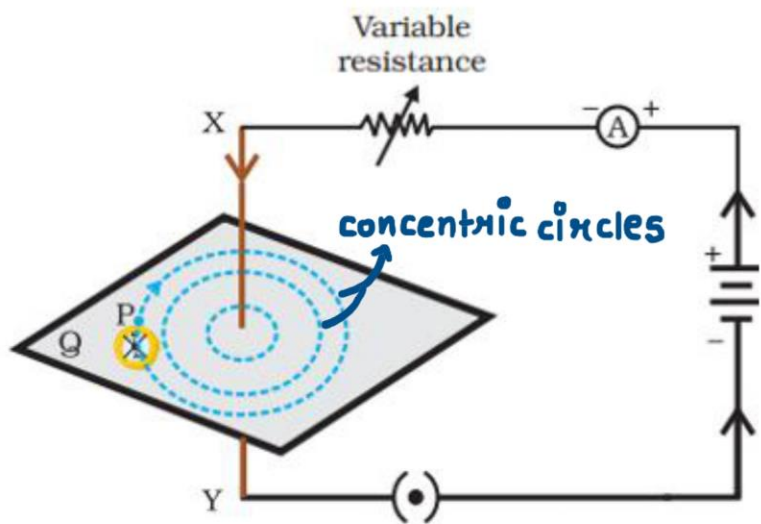
- 1 Has both direction & magnitude
outside the magnet ($N \rightarrow S$); Inside the magnet ($S \rightarrow N$)
- 2 Strength of magnetic field $\propto$ closeness of field lines
- 3 Fieldlines donot intersect $\because$ at the point of intersection compass would show 2 directions (impossible)
- 4 Direction of field lines is the direction of magnetic field; field lines make complete closed curve

UNIFORM & NON-UNIFORM MAGNETIC FIELD



OERSTED DISCOVERY: ACTIVITY

- current carrying wire produces magnetic field; its called magnetic effect of current

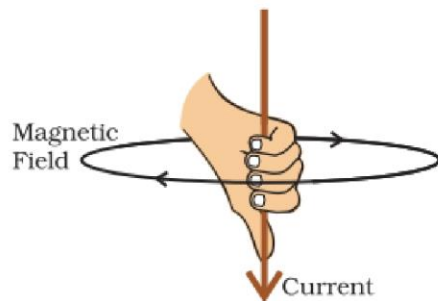


'Fe'- filings

- Max. iron filings in smallest circle ∴ near the wire
- ↑ current ∴ stronger magnetic field
- If direction of current changes; direction of mag. field also reverses
- ⊗ Inwards ⊙ outwards

Right hand rule: Maxwell

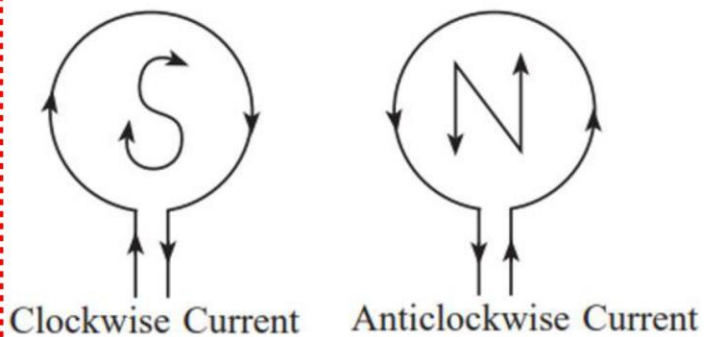
- Thumb: direction of current
- Fingers: direction of mag. field



Right-hand thumb rule

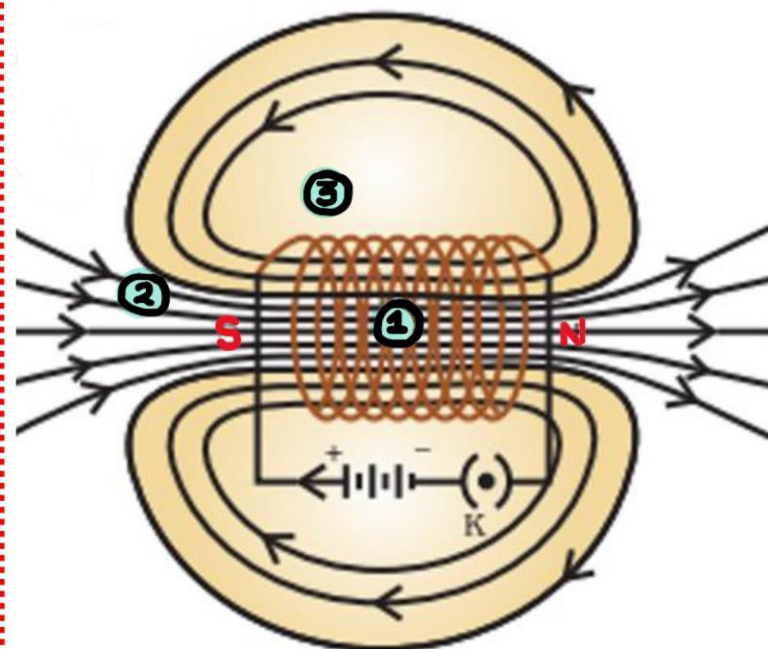
MAGNETIC FIELD DUE TO CIRCULAR LOOP

- Anticlockwise current: North pole; clockwise: south



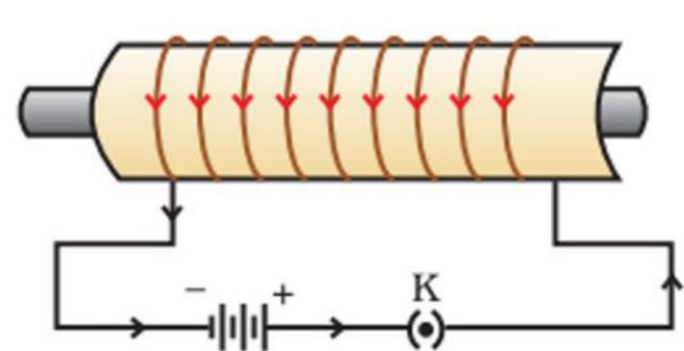
- If $I \uparrow$, then $B \uparrow$; Radius $\downarrow$, then $B \uparrow$
- no. of turns $\uparrow$ i.e., current $\uparrow$, then $B \uparrow$ (field due to each turn adds up)

MAGNETIC FIELD LINES DUE TO SOLENOID



'Cu' wire coiled in form of cylinder

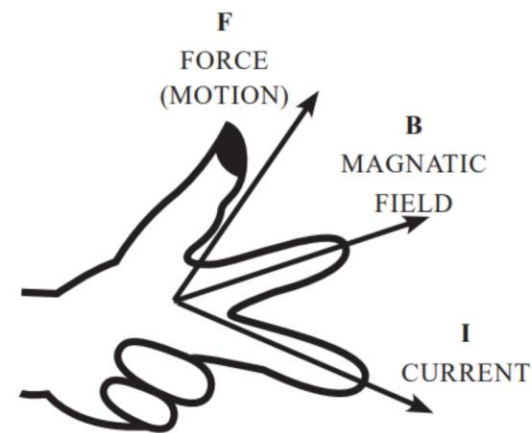
- It behaves like a bar magnet
- Uniform field inside solenoid
- $B: 1 > 2 > 3$
- $B \uparrow$ when: $I \uparrow$, turns $\uparrow$ and gap b/w turns $\downarrow$



- **Electromagnet: Temporary** - any magnetisation of magnetic materials like 'Soft Fe' inside a current carrying solenoid

FLEMING'S LEFT HAND RULE

- **Force (F)** on current carrying conductor when placed in a magnetic field



- Magnetic field can exert force on conductor which can make it move

- **F** : Force : Thumb
- **B** : Mag. field : Fore finger
- **I** : Current : Mid-finger

$$F = B i l$$

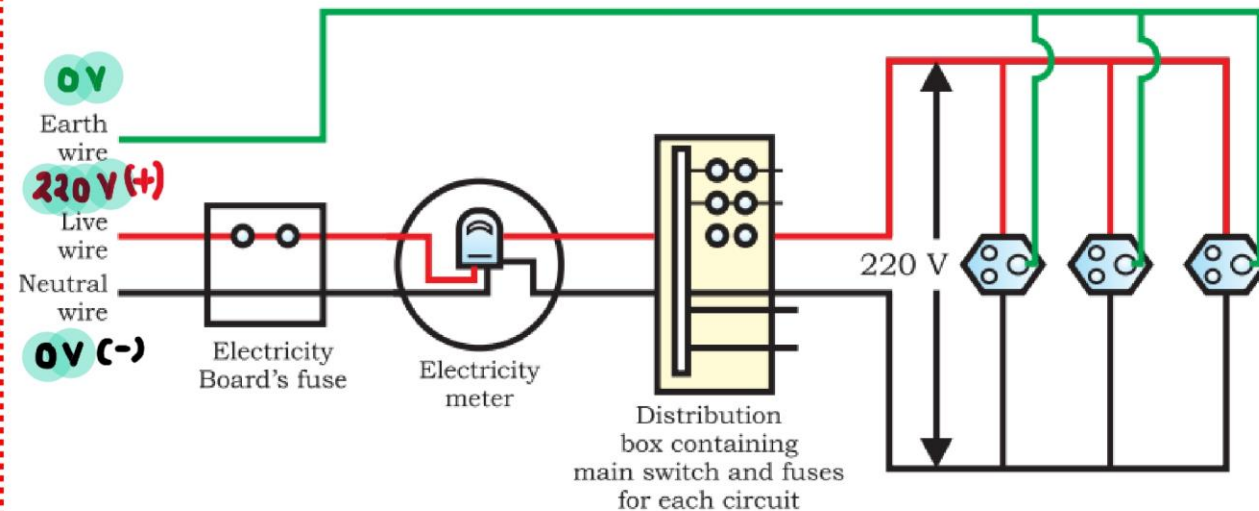
∴ $F \uparrow$ when $B \uparrow$; $F \uparrow$ when $i \uparrow$; $F \uparrow$ when length of conductor $\uparrow$

- Also depends upon angle b/w B & i

$F = \max$ (when B & i are perpendicular)

$F = \min$ (when B & i are having 0° difference) i.e, in same direction

DOMESTIC ELECTRIC CIRCUIT



- Current flow due to pot. diff. b/w neutral & live wire
- **Short circuit** : due to touching of live & neutral wire & sudden flow of large current
- **overloading** : overheating of electric wire in circuit due to large current flow (too many appliances in socket)
- ∴ Fuse used to prevent short circuit & overloading

